CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNJAB

DEPARTMENT OF INTERDISCIPLINARY COURSES IN ENGINEERING

Embedded Systems and Internet of Things Laboratory

Class Group G1 | Section A | Project Group 02

Project Abstract

Project Title: Smart Campus Environmental Monitoring System Using IoT

Abstract: This project focuses on monitoring campus environmental conditions by measuring air quality, temperature, atmospheric pressure, and rainfall using IoT technology. The system continuously gathers environmental data and processes it through an IoT-enabled microcontroller, which generates a locally accessible real-time web dashboard for instant visualization. This helps users track weather patterns, identify polluted areas, and observe changing atmospheric conditions across the campus. The solution is cost-effective, easy to deploy, and suitable for institutions aiming to promote environmental awareness and sustainability.

Application Area:

- Campus Environmental Monitoring
- Pollution Analysis and Control
- Smart and Sustainable Campuses

Impact on Society/Industry:

- Healthier Living and Working Environment: Continuous monitoring helps maintain better air quality.
- Reduction of Carbon Footprint: Helps reduce carbon footprint through data-based monitoring.
- Support for Sustainable Infrastructure: Supports sustainable and smart campus or industrial environments.

Sustainable Development Goals: SDG 3 – Good Health and Well-Being, SDG 9 – Industry, Innovation and Infrastructure, SDG 11 – Sustainable Cities and Communities, SDG 13 – Climate Action.

Technology Stack: ESP32, IoT, Environmental Sensors, Embedded C++, Cloud/IoT Platform

Group Details:

Sr. No	Name of Students	Roll No	Project Guide Name and Signature
1.	Abhay Singh Pundir	2410990006	Dr. Manvinder Sharma (ES&IOT Lab Faculty)
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Approved By:

Name of Faculty in Panel	Designation	Signature
(Lab Internal Faculty Name)	Designation of Lab Internal Faculty	

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